

5E1751

Total No. of Questions : 22

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Roll No. :

5E1751

B.Tech. V-Sem. (Main/Back) Exam. - 2024

COMPUTER SCIENCE AND ENGINEERING (IOT)

5CIT3-01 / Information Theory and Coding

CS, CCS, CIT

Time : 3 Hours

Maximum Marks : 70

Instructions to Candidates :

Attempt **all ten** questions from Part-A, **five** questions out of **seven** questions from Part-B and **three** questions out of **five** questions from Part-C.

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. Units of quantities used / calculated must be stated clearly.

*Use of following supporting material is permitted during examination.
(Mentioned in Form No. 205)*

1.

PART-A

2.

*(Answer should be given upto 25 words only. All questions are compulsory.
Each question carries 2 marks.)* [10x2=20]

Q.1. Explain the property of Mutual information.

Q.2. Prove the following statement "The sun rises in the east" contains zero amount of information.

- Q.3. Define Kraft Inequality.
- Q.4. What is a Prefix code ?
- Q.5. Compare Burst error and Random error.
- Q.6. Explain two properties of Galois field.
- Q.7. Classify the types of Linear block code.
- Q.8. What is the value of Syndrome vector for Error free transmission ?
- Q.9. Differentiate between State diagram and Trellis diagram.
- Q.10. Define Standard Array.

PART-B

(Analytical / Problem solving questions. Attempt **any five** questions.
Each question carries 4 marks.) [5x4=20]

- Q.1. Show that for a discrete binding channel :

$$H(X, Y) = H(X/Y) + H(Y)$$

$$H(X, Y) = H(X) + H(Y)$$

- Q.2. Illustrate the concept of Systematic and Non-systematic code.
- Q.3. The probabilities of five possible out come of an experiment are given as :

$$P(x_1) = 0.2, P(x_2) = 0.2, P(x_3) = 0.18, P(x_4) = 0.15, P(x_5) = 0.05$$

Determine the entropy and information rate.

- Q.4. Explain Viterbi algorithm in detail.
- Q.5. A discrete memoryless source exists four messages with probability set $[1/2, 1/4, 1/8, 1/8]$. Construct a Shannon-Fano code for this message and determine its efficiency.
- Q.6. A communication system employs a continuous source. The channel noise is White and Gaussian. The bandwidth of the source output is 10 MHz and signal to noise power ratio at the receiver is 100. What will be the channel capacity?
- Q.7. Derive the expression for channel capacity when bandwidth becomes infinite.

PART-C

(Descriptive / Analytical / Problem solving / Design questions. Attempt **any three** questions. Each question carries 10 marks.) [3x10=30]

- Q.1. Construct Haffman's code for following set of messages and determine its efficiency :

$$\begin{aligned}
 P(x_1) &= 0.49, P(x_2) = 0.14, P(x_3) = 0.14, \\
 P(x_4) &= 0.07, P(x_5) = 0.07, P(x_6) = 0.04, \\
 P(x_7) &= 0.02, P(x_8) = 0.02, P(x_9) = 0.01
 \end{aligned}$$

- Q.2. The generator polynomial of a (6, 3) cyclic code is $g(x) = 1 + x^2$:

- (a) Find all the code words for this code. [6]
- (b) How many errors can this code correct? [4]

- Q.3. Write short notes on the following :

- (a) Sequential decoding [5]
- (b) Transfer function of convolutional code [5]

Q.4. Consider a (7, 4) linear block code with the parity-check matrix H given by :

$$H = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

(a) Construct code words for (7, 4) code. [5]

(b) Show that this code is a Hamming code. [5]

Q.5. Joint probability matrix of discrete channel is given by :

$$\begin{array}{cccc} P(X, Y) = & 0.05 & 0.05 & 0.02 & 0.05 \\ & 0.15 & 0.16 & 0.01 & 0.09 \\ & 0.12 & 0.03 & 0.02 & 0.05 \\ & 0.01 & 0.12 & 0.01 & 0.06 \end{array}$$

Compute Conditional and Joint entropies.